

Question number	Answer	Additional guidance	Mark
4(a)	Choose an item. • B is the correct answer A is not the correct answer as only one statement is correct C is not the correct answer as only one statement is correct D is not the correct answer as only one statement is correct		Computer (1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	Choose an item. • B is the correct answer A is not the correct answer as P to T is not one sarcomere C is not the correct answer as Q to T is not one sarcomere D is not the correct answer as R to S is not one sarcomere		Computer (1)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	A calculation showing the following steps: • calculation of magnification (1) • answer given in standard form (1)	= 15500 (accept 15000-16000) = 1.55×10^4 accept 1.5×10^4 - 1.6×10^4 factor of 10 error 1 mark if not in standard form 15000-16000 gains 1 mark	Graduate (2)

Question number	Answer	Additional guidance	Mark
4(b)(iii)	A description that includes four of the following points: - fast twitch have fewer {capillaries / blood vessels} (1) - fast twitch have fewer mitochondria (1) - fast twitch have less myoglobin (1) - fast twitch are {larger in diameter/ longer / have more sarcoplasmic reticulum} (1) - have more {glycogen / creatine phosphate} (1)	ALLOW converse ignore less blood supply as not structural	Expert (4)

Question number	Answer	Additional guidance	Mark
4(c)	An explanation that includes the following points: • increase in velocity of contraction decreases the force of contraction (1) • (cross bridges) are formed when myosin (head) binds to {actin / myosin binding site (on actin)} / (cross bridge) are broken when ATP binds to the myosin (head) (1) • therefore reducing the total number of cross bridges (at any one time) (1)	accept slower muscle contractions have more force / converse accept inversely proportional / negative correlation accept description of a cross bridge forming accept there are more cross bridges formed in slow contraction / there are fewer cross bridges formed in fast contractions accept converse	Expert (3)

